LAB MANUAL

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (AI & ML)

Database & Management System

Submitted by-

Pooja Aggarwal

Roll no - 2401010240

Submitted to-

Mansi Kajal

Experiment – 2

Aim

To design an Entity–Relationship (ER) diagram for a student academic management system and define the relationship types among entities.

Entities and Attributes

1. Student

Represents students enrolled in the institution.

Attributes:

- User_Name (Primary Key)
- Student_Name
- Address
- Phone_Number
- Gender

2. Semester_Section

Represents a specific section offered in a semester.

Attributes:

- SSection_ID (Primary Key)
- Semester
- Section

3. Class

Represents a class in which students are enrolled.

Attributes:

- SStudentID (Primary Key)
- User_Name

4. Subject

Represents subjects offered by the institution.

Attributes:

- Subject_Code (Primary Key)
- Title
- Credits
- Semester

5. IAMarks

Represents internal assessment marks of students.

Attributes:

- SSID
- Test1
- Test2
- Test3
- FinalIA

Relationships and Relationship Types

1. Enrolls

Entities involved: Student – Class

Relationship type:

- One-to-Many (1 : N)

Explanation:
One student can enroll in multiple classes, but each class enrollment is associated with only one student.

2. Contains

Entities involved: Semester_Section – Class

Relationship type:

- One-to-Many (1 : N)

Explanation:
A semester section can contain multiple classes, but each class belongs to only one semester section.

3. Has

Entities involved: Class – IAMarks

Relationship type:

- One-to-One (1 : 1)

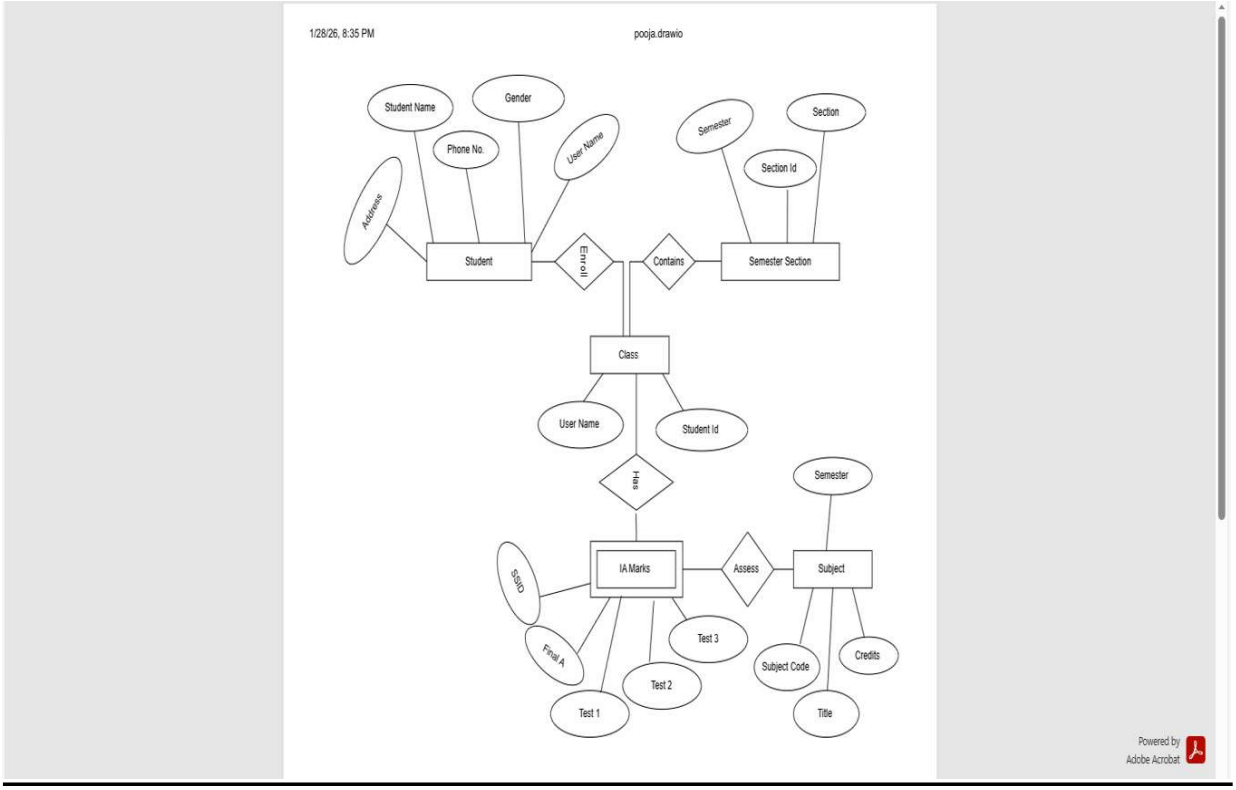
Explanation:
Each class record has one set of internal assessment marks, and each IA marks record belongs to one class.

4. Assesses

Entities involved: Subject – IAMarks

Relationship type:

- One-to-Many (1 : N)

Explanation:
A subject can assess many IA marks records for different students, but each IA marks record is related to only one subject.

Result: